MIDDLESEX SCHOOL | MATHEMATICS

Math 49 Course at a Glance

Precalculus with Limits

Review Content

REVIEW	
M49-FND-001	I can determine the domain and range of a function from a formula, graph, or table.
M49-FND-002	I can describe and apply transformations of a parent function.
M49-FND-004	I can analyze exponential functions using their equations, graphs, and contexts.
M49-FND-005	I can analyze logarithmic functions using their equations, graphs, and contexts.
M49-FND-006	I can solve exponential and logarithmic equations using equivalent forms and logarithm properties.

Function Operations, Composition, and Inverses

REQUIRED	
M49-FND-003	I can determine whether a function is even, odd, or neither.
M49-OPS-001	I can add, subtract, multiply, and divide functions and state the resulting domain.
M49-OPS-002	I can evaluate a composition from formulas, graphs, or tables.
M49-OPS-003	I can write a formula for the composition of two functions and determine its domain.
M49-OPS-004	I can decompose a function into a composition of simpler functions.
M49-OPS-005	I can determine whether a function has an inverse on a given domain.
M49-OPS-006	I can find and verify an inverse function algebraically.
M49-OPS-007	I can interpret the relationship between a function and its inverse from graphs or tables.

Power, Polynomial, and Rational Functions

REQUIRED	
M49-PWR-001	I can identify a power function and interpret the parameters in y = kx^p.
M49-PWR-002	I can construct a power function from two data points or a proportional relationship.
M49-PWR-003	I can relate the sign and size of k and p to the shape, domain behavior, and end behavior of a power function.
M49-PWR-004	I can recognize and model direct, inverse, and other power-variation relationships.
M49-PWR-005	I can solve equations involving integer or rational powers.
M49-PWR-006	I can compare the long-run dominance of logarithmic, power, polynomial, and exponential functions.
M49-PWR-008	I can analyze power functions with fractional exponents, including their domains, graph shapes, and behavior near zero.
M49-ALG-002	I can predict the end behavior of a polynomial from its degree and leading coefficient.
M49-ALG-003	I can determine polynomial zeros and their multiplicities.
M49-ALG-004	I can construct a polynomial function from specified zeros or graphical features.
M49-ALG-005	I can sketch and analyze a polynomial function using intercepts, multiplicity, and end behavior.
M49-ALG-006	I can identify holes and vertical, horizontal, or slant asymptotes of a rational function.
M49-ALG-007	I can sketch and analyze a rational function using its algebraic and asymptotic features.
M49-ALG-008	I can build or select a power, polynomial, or rational model for a context.

Limits and Function Comparison

REQUIRED	
M49-LIM-001	I can interpret limit notation as a statement about function behavior near an input.
M49-LIM-002	I can estimate a limit from a graph or table.
M49-LIM-003	I can evaluate a limit algebraically using direct substitution or simplification.
M49-LIM-005	I can interpret an infinite limit as vertical-asymptotic behavior.
M49-LIM-006	I can determine limits at infinity and relate them to long-run behavior.
M49-LIM-007	I can compare the long-run growth of polynomial, exponential, logarithmic, and rational functions.

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Sequences and Series

REQUIRED

M49-SEQ-001	I can evaluate and graph a sequence defined explicitly or recursively.
M49-SEQ-002	I can write explicit and recursive formulas for arithmetic sequences.
M49-SEQ-003	I can write explicit and recursive formulas for geometric sequences.
M49-SEQ-004	I can distinguish arithmetic, geometric, and other sequences from formulas, tables, or contexts.
M49-SEQ-005	I can create and interpret a sequence model for discrete change.
M49-SER-001	I can interpret and use summation notation.
M49-SER-002	I can calculate a finite arithmetic series.
M49-SER-003	I can calculate a finite geometric series.
M49-SER-004	I can determine whether an infinite geometric series converges and find its sum when it does.
M49-SER-005	I can create and interpret a series model for accumulated change.

Additional Limit Concepts

EXTENSION

M49-LIM-004	I can determine one-sided limits and decide whether a two-sided limit exists.
M49-LIM-008	I can distinguish a function value from a limit and determine continuity at a point.

Parametric Equations

EXTENSION

M49-PAR-001	I can graph a plane curve defined by parametric equations with its orientation.
M49-PAR-002	I can eliminate a parameter to obtain a rectangular equation.
M49-PAR-003	I can create or interpret parametric equations for motion in the plane.

Conic Sections

EXTENSION

M49-CON-001	I can classify a circle, ellipse, or hyperbola from its equation or defining geometric relationship.
M49-CON-002	I can use completing the square to rewrite a general-form conic equation in standard form.
M49-CON-003	I can write, interpret, and graph implicit and parametric equations of a circle.
M49-CON-004	I can write, interpret, and graph implicit and parametric equations of an ellipse.
M49-CON-005	I can write, interpret, and graph implicit and parametric equations of a hyperbola.
M49-CON-006	I can determine the center, vertices, foci, axes, or asymptotes needed to describe a circle, ellipse, or hyperbola.
M49-CON-007	I can explain the geometric definitions of circles, ellipses, and hyperbolas using foci and distance relationships.
M49-CON-008	I can explain a parabola as the set of points equidistant from a focus and directrix.